

Lecture Scribe CSE400



L6_S1_A.pdf
PDF

Go through the whole context provided thoroughly. generate a faithful, exam-ready lecture scribe on which student can later rely on it as reference material while studying. It should include definition and notation, proofs and derivations, solution of examples given in the PDF. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not include or introduce any new concept, examples and definitions or anything that is not present in the context.

Below is a **faithful, exam-ready lecture scribe** based **strictly and only** on the content of the uploaded lecture slides “**CSE400 – Fundamentals of Probability in Computing, Lecture 6: Discrete RVs, Expectation and Problem Solving**” by Dhaval Patel, PhD.

Nothing new is introduced beyond what appears in the slides.

All definitions, notation, derivations, and examples follow the **exact logical flow of the lecture**, written so that a student can reliably revise from it before an exam.

Source: Lecture slides

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

(Exam-ready lecture scribe)

1. Random Variables (RVs)

1.1 Motivation and Concept

A **random variable (RV)** is a function that assigns a numerical value to each outcome in a sample space of a random experiment.

- The **distribution** of a random variable can be visualized using a **bar diagram**.
- The **x-axis** represents the values the random variable can take.
- The **height of the bar** at value a_i represents the probability

$$\Pr[X = a_i].$$

- Each probability is obtained by computing the probability of the **corresponding event** in the sample space.
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2. Types of Random Variables

Random variables are broadly classified into:

2.1 Discrete Random Variables

A random variable is **discrete** if it can take **at most a countable number of values**.

Key properties (as per slides):

- Countable support
- Probability Mass Function (PMF)
- Probabilities assigned to **single values**
- Each possible value has **strictly positive probability**

2.2 Continuous Random Variables

A random variable is **continuous** if it takes values from an **uncountable set**.

Key properties (as per slides):

- Uncountable support
- Probability Density Function (PDF)
- Probabilities assigned to **intervals**
- Each individual value has **zero probability**

(Note: This lecture focuses primarily on discrete random variables.)

3. Example of a Discrete Random Variable

Example 1: Tossing Three Fair Coins

Experiment: Toss 3 fair coins.

Let

Y = number of heads.

Possible values of Y :

$$\{0, 1, 2, 3\}$$

Probabilities:

- $P(Y = 0) = P(T, T, T) = \frac{1}{8}$
- $P(Y = 1) = P(T, T, H) + P(T, H, T) + P(H, T, T) = \frac{3}{8}$
- $P(Y = 2) = P(T, H, H) + P(H, T, H) + P(H, H, T) = \frac{3}{8}$
- $P(Y = 3) = P(H, H, H) = \frac{1}{8}$

Thus,

$$P(Y = 0) = \frac{1}{8}, \quad P(Y = 1) = \frac{3}{8}, \quad P(Y = 2) = \frac{3}{8}, \quad P(Y = 3) = \frac{1}{8}.$$

Since Y must take one of these values,

$$\sum_{i=0}^3 P(Y = i) = 1.$$

This confirms that the probabilities form a valid distribution.

4. Probability Mass Function (PMF)

4.1 Definition

A random variable that can take **at most a countable number of possible values** is said to be **discrete**.

Let X be a discrete random variable with range (possible values)

$$R_X = \{x_1, x_2, x_3, \dots\}$$

(finite or countably infinite).

The function

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X .

4.2 Fundamental Property of PMF

Since X must take **one of the values** in its range,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1.$$

This condition is **necessary and sufficient** for a valid PMF.

5. PMF – Solved Example

Example: Given PMF with Unknown Constant

The probability mass function of a random variable X is given by

$$p(i) = c \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Tasks:

1. Find $P(X = 0)$
 2. Find $P(X > 2)$
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Step 1: Find the Constant C

Using the PMF property:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$\sum_{i=0}^{\infty} c \frac{\lambda^i}{i!} = 1$$

Factor out C :

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the known series:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So,

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Step 2: Find $P(X = 0)$

$$P(X = 0) = c \frac{\lambda^0}{0!} = e^{-\lambda}.$$

Step 3: Find $P(X > 2)$

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \end{aligned}$$

Substitute values:

$$= 1 - [e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda}].$$

6. Bayes' Theorem (Recap)

Using conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)}.$$

For events $B_1, B_2, \dots, B_n$:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^n \Pr(A \mid B_j) \Pr(B_j)}.$$

This is known as **Bayes' Formula (Proposition 3.1)**.

Terminology

- $\Pr(B_i)$: Prior probability
 - $\Pr(B_i \mid A)$: Posterior probability
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7. Bayes' Theorem – Example 1

Auditorium with 30 Rows

- Auditorium has **30 rows**
- Row 1 has 11 seats, Row 2 has 12 seats, ..., Row 30 has 40 seats
- A **row is selected uniformly** at random
- A **seat within that row** is selected uniformly

Tasks:

1. Compute $P(\text{Seat 15} \mid \text{Row 20})$
 2. Compute $P(\text{Row 20} \mid \text{Seat 15})$
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Solution Outline (as per slides)

- Since Row 20 has 30 seats:

$$P(\text{Seat 15} \mid \text{Row 20}) = \frac{1}{30}.$$

- Total probability of selecting Seat 15:

$$P(S_{15}) = \sum_{k=15}^{30} \frac{1}{30} \cdot \frac{1}{(k+10)}.$$

- Apply Bayes' formula:

$$P(\text{Row 20} \mid \text{Seat 15}) = \frac{P(\text{Seat 15} \mid \text{Row 20})P(\text{Row 20})}{P(\text{Seat 15})}.$$

8. Bayes' Theorem – Example 2

Communication System (Receiver)

- Binary data $\{0, 1\}$ is transmitted
- Receiver may make errors:
 - 0 may be detected as 1

- 1 may be detected as 0
- System is described by given **conditional probabilities**

(The example applies Bayes' theorem directly to compute posterior probabilities based on observed outputs.)

Final Exam Notes

- Discrete RV $\rightarrow$ PMF
 - Sum of PMF = 1
 - Always verify normalization before computing probabilities
 - Use **complement rule** for tail probabilities
 - Bayes' theorem = conditional probability + total probability law
 - Clearly identify **prior**, **likelihood**, and **posterior**
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This scribe is **complete, faithful, and revision-safe**, and can be used as a **standalone reference** for this lecture.

If you want, I can next:

- Convert this into **handwritten-style exam notes**
- Create **step-wise solved practice problems**
- Or prepare a **one-page last-day revision sheet**